Answer key

SAT Practice Test 5, Reading and Writing

Module 1

NO.	ANSWER	SKILL
1	**D**	Words in context
2	**A**	Words in context
3	**B**	Words in context
4	**C**	Words in context
5	**B**	Text structure and purpose
6	**A**	Text structure and purpose
7	**A**	Central ideas and details
8	**C**	Central ideas and details
9	**D**	Cross-text connections
10	**B**	Command of evidence: quantitative
11	**B**	Command of evidence: quantitative
12	**C**	Command of evidence: textual
13	**D**	Command of evidence: textual
14	**C**	Inferences
15	**B**	Inferences
16	**A**	Boundaries
17	**C**	Form, structure, and sense
18	**D**	Form, structure, and sense
19	**A**	Form, structure, and sense
20	**B**	Boundaries
21	**D**	Boundaries
22	**A**	Transitions
23	**A**	Transitions
24	**B**	Transitions
25	**C**	Rhetorical synthesis
26	**D**	Rhetorical synthesis
27	**A**	Rhetorical synthesis

Module 2

NO.	ANSWER	SKILL
1	**D**	Words in context
2	**A**	Words in context
3	**C**	Words in context
4	**B**	Words in context
5	**A**	Text structure and purpose
6	**D**	Text structure and purpose
7	**D**	Cross-text connections
8	**C**	Central ideas and details
9	**A**	Central ideas and details
10	**B**	Command of evidence: quantitative
11	**B**	Command of evidence: quantitative
12	**A**	Command of evidence: textual
13	**C**	Command of evidence: textual
14	**C**	Inferences
15	**D**	Inferences
16	**A**	Boundaries
17	**C**	Form, structure, and sense
18	**A**	Boundaries
19	**D**	Form, structure, and sense
20	**B**	Form, structure, and sense
21	**A**	Boundaries
22	**B**	Transitions
23	**A**	Transitions
24	**D**	Transitions
25	**B**	Rhetorical synthesis
26	**C**	Rhetorical synthesis
27	**D**	Rhetorical synthesis

Answer key

SAT Practice Test 5, Math

Module 1

NO.	ANSWER	SKILL
1	**C**	Data analysis
2	**A**	Linear equations
3	**D**	Linear equations in context
4	**B**	Systems of equations
5	**B**	Functions
6	**C**	Nonlinear expressions
7	**D**	Triangles
8	**D**	Probability
9	**C**	Scatterplots
10	**D**	Inequalities
11	**A**	Functions
12	**B**	Exponential functions
13	**C**	Linear functions
14	**D**	Percent change
15	**A**	Volume
16	**B**	Right triangles
17	**10**	Quadratics
18	**2**	Linear equations
19	**25**	Systems of equations
20	**15**	Right triangles
21	**15**	Percent change
22	**396**	Volume

Module 2

NO.	ANSWER	SKILL
1	**D**	Percentages
2	**A**	Volume
3	**C**	Inequalities
4	**B**	Statistics
5	**A**	Functions
6	**B**	Probability
7	**D**	Systems of equations
8	**B**	Statistics
9	**C**	Systems of equations
10	**A**	Linear functions in context
11	**B**	Surface area
12	**D**	Functions
13	**D**	Exponential equations
14	**A**	Percentages
15	**C**	Exponential functions
16	**B**	Equivalent expressions
17	**28**	Systems of equations
18	**36**	Quadratics
19	**60**	Similar triangles
20	**7**	Circles in the plane
21	**6**	Quadratics
22	**5**	Systems with nonlinear equations

Score yourself. Count the questions you got right in each module.

Reading and Writing, Module 1: ___ of 27

Module 2: ___ of 27

Math, Module 1: ___ of 22

Module 2: ___ of 22

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 1

Questions 1 to 14

1 **Choice D** **Words in context**

A trail far steeper than expected would dismay hikers. Delighted and relieved suggest pleasure, and indifferent does not fit the contrast with the map.

2 **Choice A** **Words in context**

The colon explains that stone survives thousands of years, which is durability. Fragile is the opposite, and the passage does not mention price or portability.

3 **Choice B** **Words in context**

Her cheerful manner contrasted with the grim news, so it belied it. Underscored and reflected would match the seriousness.

4 **Choice C** **Words in context**

Though sets up a contrast with the later criticism, so the first volume was received with acclaim. Scorn would not contrast.

5 **Choice B** **Text structure and purpose**

The underlined sentence gives examples of different flash patterns, supporting the claim that patterns are specific to each species. It does not dispute the claim, or discuss females not flashing or food.

6 **Choice A** **Text structure and purpose**

The speech weighs on Isidore like a stone, but at the end he realises he does not need it, which shows a shift from anxiety to confidence. The passage does not criticise the guests, and he is not unprepared.

7 **Choice A** **Central ideas and details**

The text reports that much of the benefit seemed to be relief of withdrawal. It does not say decaffeinated coffee is better, that drinkers always know their dose, or that caffeine has no effect on anyone.

8 **Choice C** **Central ideas and details**

The text says English borrowed words with foods but that words also travel in the other direction. It makes no claim of most borrowings, about how often foods are named, or that borrowed words are never changed.

9 **Choice D** **Cross-text connections**

Text 2 says height limits raise prices by limiting supply and proposes design rules instead. The economist does not ignore character or deny that low-rise streets encourage walking.

10 **Choice B** **Command of evidence: quantitative**

Fewer than eight hours includes the 5, 6 and 7 hour groups: 30 + 75 + 105 = 210 students, or 70 percent of 300. The other choices describe the opposite group or a single bar.

11 **Choice B** **Command of evidence: quantitative**

A claim about every district needs data covering all four, and the first choice states that complaints fell in all while the table shows this. The others cover only some districts or contradict the claim.

12 **Choice C** **Command of evidence: textual**

The claim links reduced food to exploration, which the quotation about a few ants leaving the trail shows. The others describe the study, the following behaviour or her interpretation.

13 **Choice D** **Command of evidence: textual**

Longer endurance with music is the direct support. The others describe the design.

14 **Choice C** **Inferences**

Fenced plots lost species, so the elephants' disturbance may maintain diversity. The other choices contradict the findings.

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 1

Questions 15 to 27

15 Choice B Inferences

Customers came for the owner's recommendations, and the shop closed when the owner died, so success depended on the owner's knowledge and relationships. The other options are not supported.

16 Choice A Boundaries

The relative clause which lasted nearly an hour is closed by a comma before the main verb left.

17 Choice C Form, structure, and sense

Data is conventionally treated as plural in formal writing, so show agrees.

18 Choice D Form, structure, and sense

The items in the list must be parallel nouns, so violin, piano, and singing is correct.

19 Choice A Form, structure, and sense

The possessive their is needed. There and they're are other words, and its does not agree with the plural hikers.

20 Choice B Boundaries

The pronoun is the subject of oversaw, so who is the right case, and the clause is complete with the verb oversaw.

21 Choice D Boundaries

The second clause contrasts with thoroughness, so nevertheless is correct.

22 Choice A Transitions

The slow service contrasts with the excellent food, so However is correct.

23 Choice A Transitions

Over time describes what the islanders learned across generations of fishing, so it is the logical transition.

24 Choice B Transitions

The reports contradict the promise, so Yet is the right transition.

25 Choice C Rhetorical synthesis

To explain the event to a beginner, the choice should give where it began, how long it burned and its toll, which the first does.

26 Choice D Rhetorical synthesis

A contrast needs the properties of both materials in terms of the same measure, which the first choice gives.

27 Choice A Rhetorical synthesis

Scale is shown by comparing the small group with the huge number of saplings, which the first choice does.

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 2

Questions 1 to 14

1 **Choice D** Words in context

Fear of censure suggests the conclusions clashed with the orthodoxy, so at odds fits. In step would give no reason for secrecy.

2 **Choice A** Words in context

The surprise at the author's frankness about failures implies that critics expected a boastful, self-congratulatory memoir. Unsparing would match the candour, so it is the opposite of what critics expected.

3 **Choice C** Words in context

A gift that transformed the collection is a turning point, which watershed means. Setback and digression imply no improvement.

4 **Choice B** Words in context

Contradictory statements make an account inconsistent. Lucid, corroborated and unassailable suggest strength or support.

5 **Choice A** Text structure and purpose

The underlined sentence shows the problem was serious enough for a large prize, and the next sentence says the solution came from a carpenter. It does not explain latitude, blame the government or describe training.

6 **Choice D** Text structure and purpose

Selwyn lets the boy fail in private and nods toward the shore as if nothing had gone wrong, which shows tact and patience. He is not mocking the boy, and the passage does not suggest reluctance to learn or discuss his fishing.

7 **Choice D** Cross-text connections

Text 2 gives a different explanation, pattern detection with plentiful exposure, for the fast learning that Text 1 attributes to inborn grammar. The researcher does not deny fast learning.

8 **Choice C** Central ideas and details

The text reports ongoing neuron formation and a possible role in keeping memories distinct. The rate declines, the role is still debated, and blocking the neurons did affect memory.

9 **Choice A** Central ideas and details

The text says X-rays show that the artist first painted a tense figure and then revised it to create calm. It does not say the work is a forgery, that he kept the first idea, or that its colours are admired.

10 **Choice B** Command of evidence: quantitative

The claim is that the gap was largest at the first attempt, which the first choice shows with the gap and its comparison. The others cover one later attempt, a general statement or a single figure.

11 **Choice B** Command of evidence: quantitative

The claim is about both smaller reserves, which the first choice covers with correct figures. The others discuss the larger reserves, one reserve, or the total.

12 **Choice A** Command of evidence: textual

The claim ties the clean manuscripts to his method, and only the quotation describing slow composition and setting the sentence down when finished does so. The others state the result or who reported it.

13 **Choice C** Command of evidence: textual

Only the combined exposure produced an effect, while each chemical alone had none. This is stated in full only in the first choice. The others give half of the pattern.

14 **Choice C** Inferences

Scattered guesses averaged to nearly the true number, so the errors canceled out. The other choices contradict the result.

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 2

Questions 15 to 27

15 **Choice D** **Inferences**

Simultaneous independent invention suggests shared conditions made the idea available to many people. The other choices are not supported.

16 **Choice A** **Boundaries**

The subject of the clause is the plural novels, which, so the plural have been is correct.

17 **Choice C** **Form, structure, and sense**

The subject director is singular, and the past tense attended agrees and fits the context.

18 **Choice A** **Boundaries**

After Not until, the auxiliary comes before the subject, and the singular cause needs was.

19 **Choice D** **Form, structure, and sense**

The list should be parallel noun phrases: careful planning, reliable equipment, and the cooperation of local guides.

20 **Choice B** **Form, structure, and sense**

The pronoun is the subject of had been estranged, referring to people, so who is correct.

21 **Choice A** **Boundaries**

A colon introduces the explanation of how the members were divided, after a complete clause. A comma would create a comma splice, and the other choices would leave the sentence ungrammatical.

22 **Choice B** **Transitions**

The sentence corrects the assumption by supplying the fact, so In fact is the best transition.

23 **Choice A** **Transitions**

The sentence describes how reception changed over time, so Over the years is the logical transition.

24 **Choice D** **Transitions**

The storm is an unfortunate event that contradicts the planning, so Unfortunately is the best transition.

25 **Choice B** **Rhetorical synthesis**

The unusual behaviour is the extraordinary memory for hidden seeds, which only the first choice states.

26 **Choice C** **Rhetorical synthesis**

Significance is expressed by the credited influence on conservation, which the first choice states along with the book's identity.

27 **Choice D** **Rhetorical synthesis**

The cause and effect are the rain, the rising river and the closed road, which the first choice links. The others pair unrelated facts.

Answer explanations

SAT Practice Test 5, Math, Module 1

Questions 1 to 11

1 **Choice C** Data analysis

Team 3 has 42 and Team 2 has 9, and 42 minus 9 is 33.

2 **Choice A** Linear equations

Subtract 3x from both sides: 3x + 7 = 22. Then 3x = 15, so x = 5.

3 **Choice D** Linear equations in context

The fee is paid once, so it is the constant 27. The $23 repeats, so it multiplies m. The total is 23m + 27.

4 **Choice B** Systems of equations

Adding the equations gives 2x = 8, so x = 4.

5 **Choice B** Functions

Each step of 1 in x adds 2 to f(x), so the slope is 2. Then f(1) = 8 gives 2 + 6, so the intercept is 6.

6 **Choice C** Nonlinear expressions

Multiply the coefficients, 3 times 4 is 12, and add the exponents, 5 plus 3 is 8.

7 **Choice D** Triangles

The angles of a triangle add to 180°, so 2x + 3x + 4x = 9x = 180 and x = 20.

8 **Choice D** Probability

39 students travel that way and 19 of them are in Grade 10, so the probability is 19/39. The fraction with 134 on the bottom uses every student.

9 **Choice C** Scatterplots

Substitute x = 8: 4(8) + 39 = 32 + 39 = 71.

10 **Choice D** Inequalities

The total is the 44 already joined plus 25 each week, and it must be at least 120, so 44 + 25w ≥ 120.

11 **Choice A** Functions

f(5) = 3(25) − 10 + 4 = 75 − 10 + 4 = 69.

Answer explanations

SAT Practice Test 5, Math, Module 1

Questions 12 to 22

12 **Choice B** Exponential functions

Tripling every 6 hours means the number of triplings is t/6, so the population is 800 times 3 to the power t/6.

13 **Choice C** Linear functions

The slope is (18 − 12)/(6 − 3) = 2. Using (3, 12), 12 = 2(3) + b gives b = 6.

14 **Choice D** Percent change

The increase is 900, and dividing by the original 6,000 gives 0.15, which is 15%. Dividing by the new value would give a smaller percent.

15 **Choice A** Volume

The volume is $\pi r^2 h = \pi(16)(12) = 192\pi$.

16 **Choice B** Right triangles

$7^2 + 24^2 = 625$, so the hypotenuse is $\sqrt{625} = 25$.

17 **10** Quadratics

The equation factors as (x − 10)(x + 2) = 0, so the solutions are 10 and −2. The positive one is 10.

18 **2** Linear equations

Expanding gives 5x − 25 = 6x − 27. Collecting terms gives −x = −2, so x = 2.

19 **25** Systems of equations

Adding x + y = 30 and x − y = 20 gives 2x = 50, so the larger number is 25.

20 **15** Right triangles

By the Pythagorean theorem, the other leg is $\sqrt{289 - 64} = \sqrt{225} = 15$.

21 **15** Percent change

The increase is 12 dollars, and 12 divided by the original 80 is 0.15, which is 15 percent.

22 **396** Volume

The volume is $\pi r^2 h = \pi(36)(11) = 396\pi$, so k = 396.

Answer explanations

SAT Practice Test 5, Math, Module 2

Questions 1 to 11

1 **Choice D** Percentages

40% of 300 is 0.4 times 300, which is 120.

2 **Choice A** Volume

The volume is $\pi r^2 h = \pi(9)(6) = 54\pi$.

3 **Choice C** Inequalities

The total is the 54 already joined plus 15 each week, and it must be at least 200, so $54 + 15w \geq 200$.

4 **Choice B** Statistics

The first 5 scores total 310, and adding 80 gives 390. Divided by 6, the mean is 65.

5 **Choice A** Functions

$f(4) = 4(16) - 8 + 4 = 64 - 8 + 4 = 60$.

6 **Choice B** Probability

40 students travel that way and 19 of them are in Grade 10, so the probability is 19/40. The fraction with 123 on the bottom uses every student.

7 **Choice D** Systems of equations

Adding the equations gives $2x = 28$, so $x = 14$.

8 **Choice B** Statistics

Multiplying by 2 multiplies the mean and the standard deviation by 2; adding 11 moves the mean by 11 but does not change the spread. The mean of A is 6, so the mean of B is 23, which is 17 greater. The standard deviation of B is 2 times that of A.

9 **Choice C** Systems of equations

A system has no solution when the lines are parallel but not identical. The second equation is 3 times the left side of the first with a different constant, so a must equal 6/3 = 2.

10 **Choice A** Linear functions in context

The coefficient of x is the change in cost for each extra hour. The constant 30 is the fixed fee, and the 1 hour cost would be 34.2.

11 **Choice B** Surface area

Each cube has 6 faces, and gluing two faces together hides 2 of the 12, leaving 10 faces of area s^2. So $10s^2 = 640$, $s^2 = 64$, and $s = 8$.

Answer explanations

SAT Practice Test 5, Math, Module 2

Questions 12 to 22

12 **Choice D** Functions

g(2) = 7, so f(g(2)) = f(7) = 34. Also f(2) = 14, so g(f(2)) = g(14) = 199. The difference is 34 − 199 = −165.

13 **Choice D** Exponential equations

Write 4 as 2^2, so the right side is 2^(2x − 2). Equal bases give x + 4 = 2x − 2, so x = 6.

14 **Choice A** Percentages

Start at 100. After the 50% rise the price is 150, and 20% off 150 leaves 120. That is 20% higher than the original.

15 **Choice C** Exponential functions

A year is 12 months, so the value is multiplied by 0.9^(12/6) = 0.9^2 = 0.81 each year. That is a decrease of 1 − 0.81, or 19%.

16 **Choice B** Equivalent expressions

$(3x + 2)^2 = 9x^2 + 12x + 4$, and $(x − 4)^2 = x^2 − 8x + 16$. Subtracting the second from the first gives $8x^2 + 20x − 12$.

17 **28** Systems of equations

Adding x + y = 32 and x − y = 24 gives 2x = 56, so the larger number is 28.

18 **36** Quadratics

The maximum is at t = −b/(2a) = 8/2 = 4. Then h(4) = −1(16) + 32 + 20 = 36.

19 **60** Similar triangles

The scale factor is DE / AB = 12 / 3 = 4. The perimeter of ABC is 15, so the perimeter of DEF is 4 × 15 = 60.

20 **7** Circles in the plane

Complete the square: $(x + 2)^2 + (y + 6)^2 = 9 + 4 + 36 = 49$. The radius is √49 = 7.

21 **6** Quadratics

Exactly one real solution means the discriminant is zero: $b^2 − 4(9) = 0$, so $b^2 = 36$ and b = 6 (taking the positive value).

22 **5** Systems with nonlinear equations

Setting the expressions equal gives $x^2 − 4x + (9 − k) = 0$. One solution means the discriminant is 0: 16 − 4(9 − k) = 0, so k = 5.